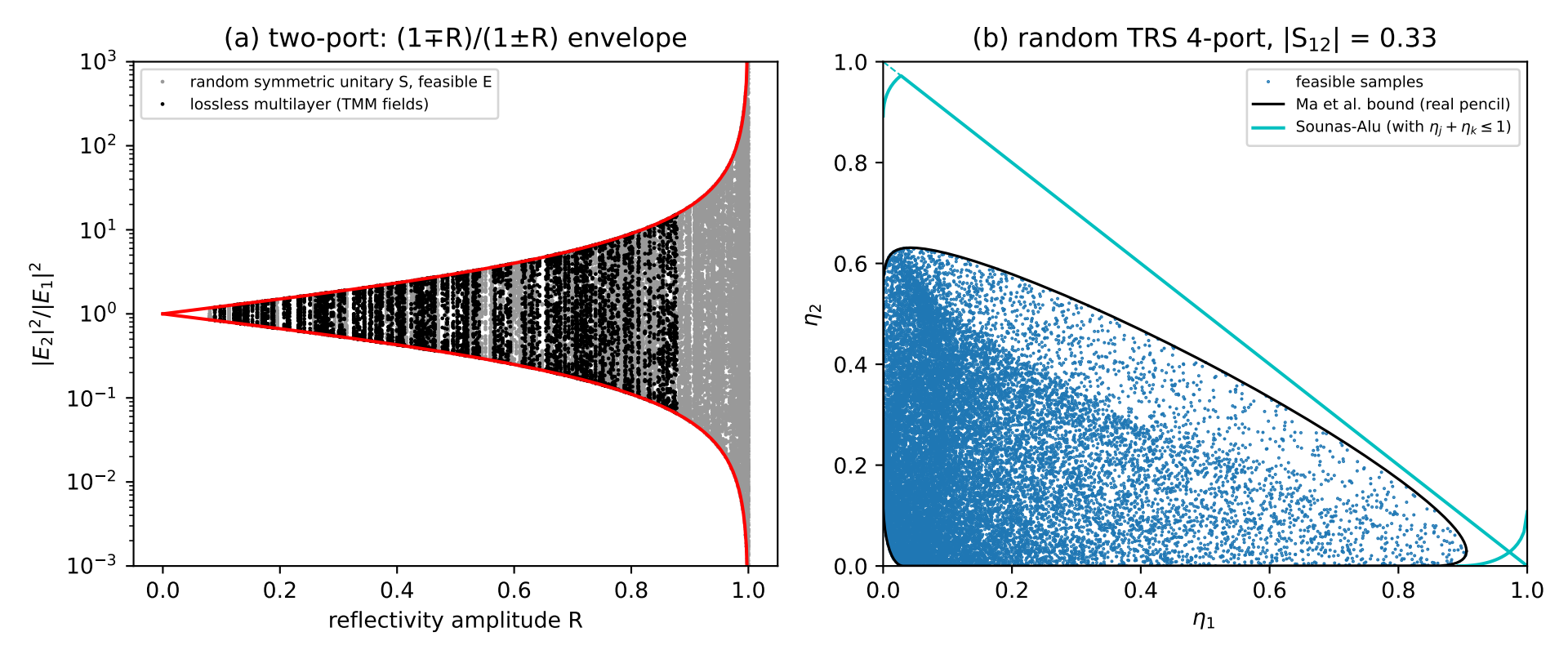

(a) two-port: (1∓R)/(1±R) envelope
random symmetric unitary S, feasible E
lossless multilayer (TMM fields)
$|E_2|^2/|E_1|^2$
reflectivity amplitude R
(b) random TRS 4-port, $|S_{12}|$ = 0.33
feasible samples
Ma et al. bound (real pencil)
Sounas-Alu (with $\eta_j + \eta_k \leq 1$)
η_2
η_1